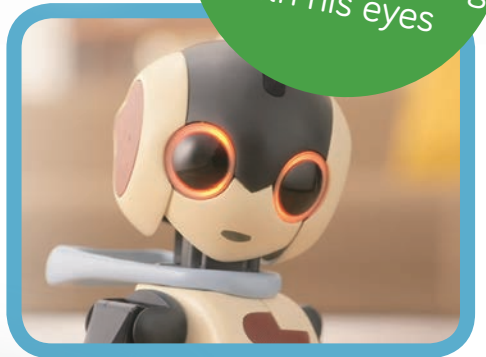
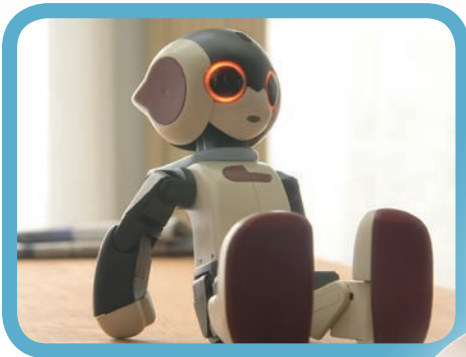
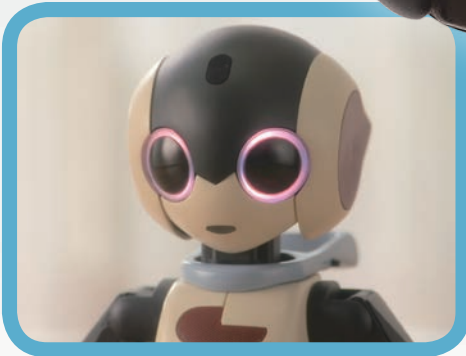


Build your own Robi

Pack **01**



Stages 1-2:
Begin assembling
Robi's head, starting
with his eyes



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Your step-by-step guide to building Robi

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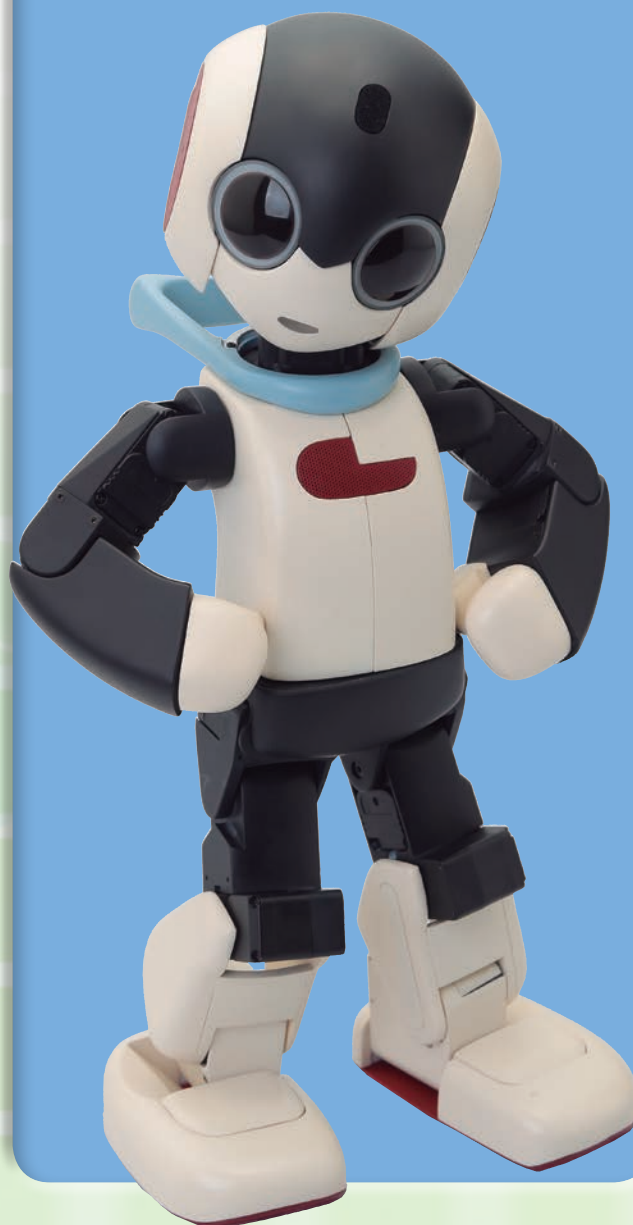
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STAGE 1: PREPARING ROBI'S EYES

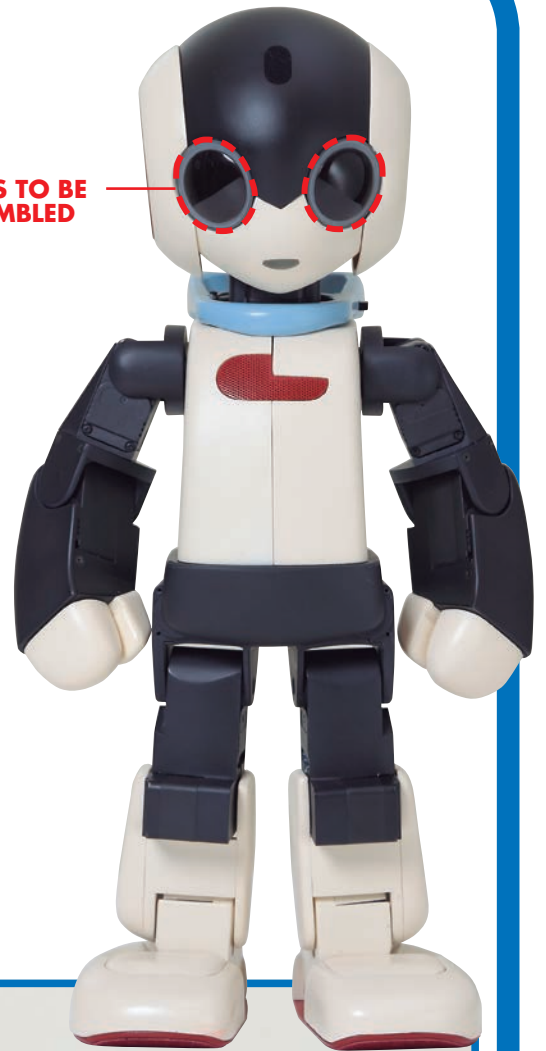
The first thing you will do is to start putting together Robi's eyes. This simply involves wrapping the black centres with reflective tape, and then fitting the translucent white borders.

Robi expresses his different states of emotion and activity by changing the colour of his eyes. As the central part of each eye is a deep black (this is to conceal a sensor that you will fit later on), it is the translucent white border that lights up when illuminated from inside by coloured LEDs (also fitted at a later stage). To diffuse the light evenly, you need to cover the black parts of Robi's eyes with reflective tape

before fitting the borders. This isn't difficult, but it needs to be done neatly to avoid wrinkling the self-adhesive silver tape, so go slowly and carefully.

In addition to your first parts, you have also received a 'mini Robi' figurine. This is a scale model of the full-size Robi you will be building over the coming stages, giving you an idea of his overall proportions and where the parts fit.

PARTS TO BE
ASSEMBLED



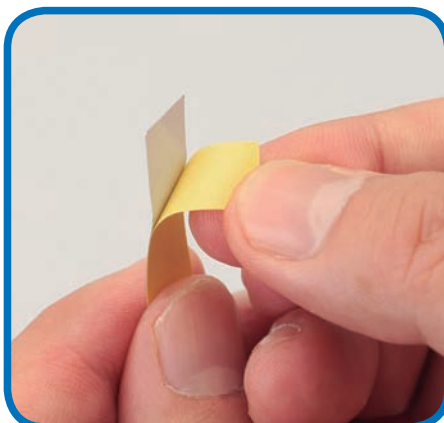
YOUR PARTS

- 1 Right ear
- 2 Right ear trim panel
- 3 Top of head
- 4 Eyes (x 2)
- 5 White borders (x 2)
- 6 Reflective tapes (x 2)

ADDING REFLECTIVE TAPE



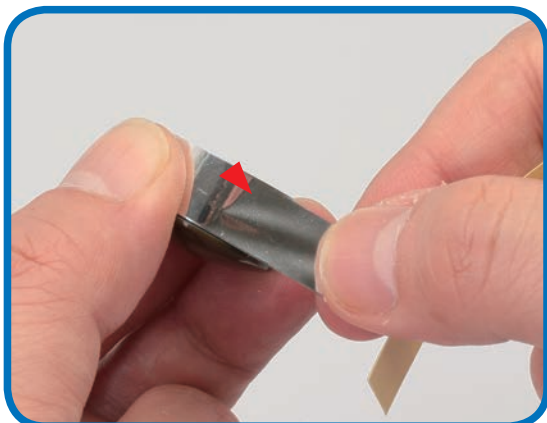
1 Peel the protective plastic film off one of the black eyes.



2 Pull one end of the backing paper off a strip of reflective tape to expose about 3cm of the adhesive.



3 Very carefully, begin to wrap the reflective tape around the eye, so one edge fits just under the rim.



4 Keeping the tape taut to prevent any creases forming, gradually pull away the remaining backing paper and smooth the adhesive side down with your thumb.



5 Work slowly and carefully around the eye until you reach your starting point. Overlap the end by about 1mm and smooth it firmly onto the surface.

Smooth the tape with your thumb from front to back to remove any wrinkles or creases.



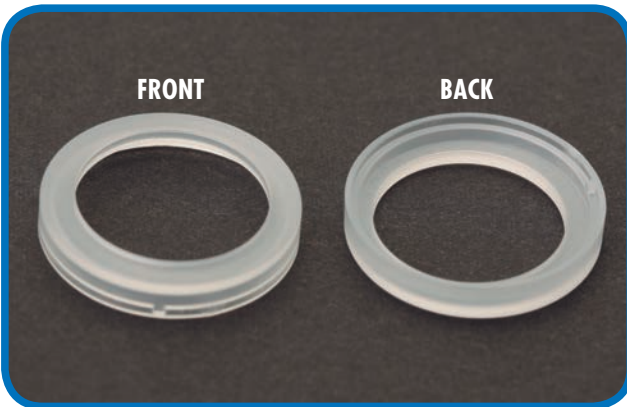
6 Repeat Steps 1 to 5 for the second eye.



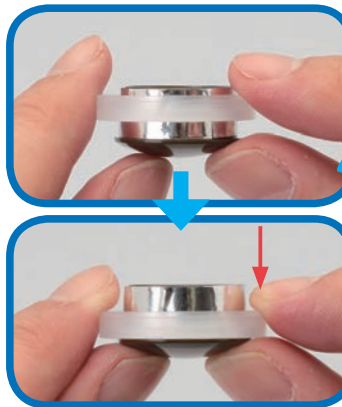
7 After taping both eyes, they should look like this.

ADDING THE WHITE BORDERS

BE CAREFUL!



- 8 Compare the white borders, as shown above, so you can see which way round they fit. The front is raised and the back has a recessed rim.



- 9 Slide a white border over one of the black eyes, with the raised front of the border on the same side as the rounded front of the eye. Be sure to keep the parts straight.



Do not force the border on at an angle. This is harder to do, and may crease or tear the reflective tape.

Assembled eyes



Repeat Step 9 for the second eye to complete this stage.

KEEP THE PARTS TIDY



Store the unused parts away safely in plastic bags until next time. This will protect the polished surfaces and prevent them from becoming scratched.

A FACE THAT LIGHTS UP!



Later on in the assembly, you will be fitting Robi's eyes with LEDs, which will illuminate the white borders you have just fitted. The LEDs come in a range of colours including orange, green, pink, purple and blue. The colour varies depending on Robi's mood and the activity he is currently performing: for example, when dancing, his eyes will continuously change colour (as well as flash on and off) to the music. Robi's mouth is also fitted with an LED that lights up when he is talking, further adding to his expressive qualities.



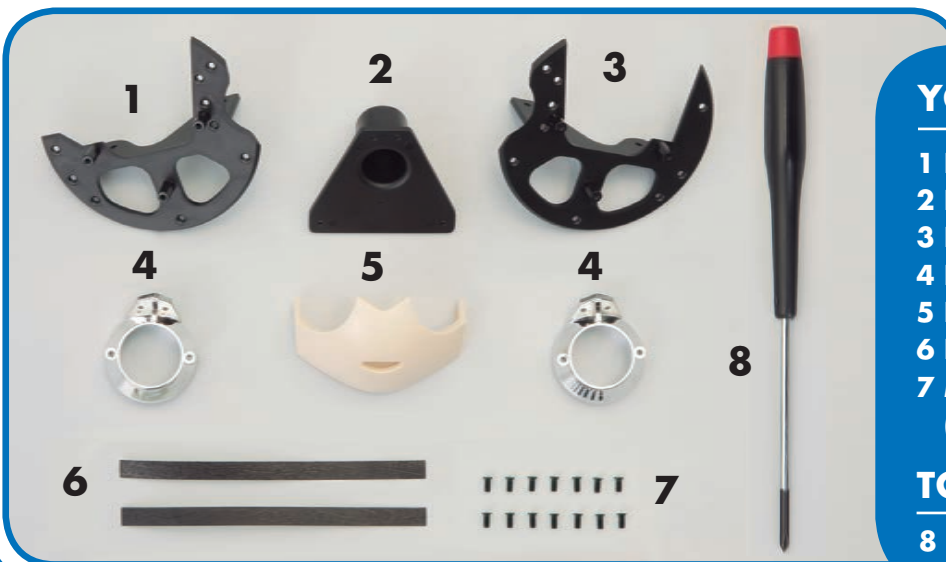
STAGE 2: START PUTTING TOGETHER ROBI'S HEAD

Using the screws provided, assemble the framework for Robi's head before fitting his eyes and face, after which Robi's characteristic features will start to take shape.

It is not difficult to put the head together, but note that the two side frames look similar, so make sure you don't get them confused. Use the specially designed screwdriver provided with this stage, holding it as

shown in the 'Close-Up' box below. To complete the assembly you will need the two eyes that you put together in Stage 1, so have these to hand, and make sure you tape them neatly at Steps 12-14.

PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Left head frame
- 2 Neck mounting
- 3 Right head frame
- 4 Eye sockets (x 2)
- 5 Face
- 6 Black tape (100mm strips x 2)
- 7 M2 x 4.5mm countersunk screws (x 14 – includes 2 spares)

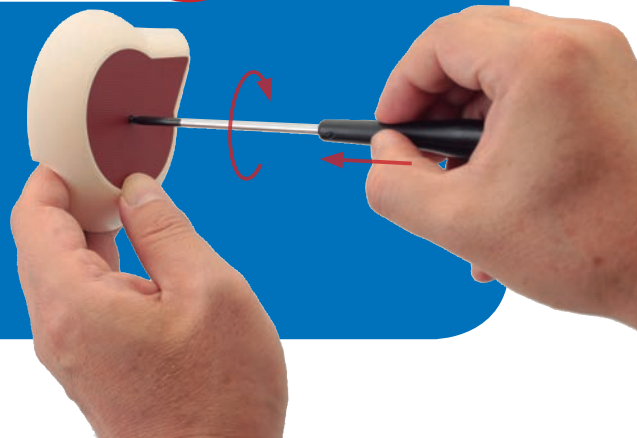
TOOLS

- 8 Crosshead screwdriver

YOUR SPECIAL SCREWDRIVER



The only special tool you need to assemble Robi is the screwdriver that you received with this stage. The tip is specially designed to fit the small screws that hold Robi together, and the end of the handle has a revolving pad which is designed to help you apply even pressure with your palm. This will keep the screw in place while you turn the handle with your thumb, index and middle fingers. Never grip the handle itself like a conventional screwdriver. Turning the screws as shown makes it much easier to put them in straight and reduces the chance of overtightening, which can damage both the screw and the parts being joined together.



ASSEMBLING THE FRAMES



1 Before you begin, sort out the left and right head frames. Lay both of them out on your work surface, facing each other as shown. Compare them with the photo to ensure that you can tell the left and right frames apart. The two are almost symmetrical, but the back of the left frame is shorter than the equivalent area on the right.

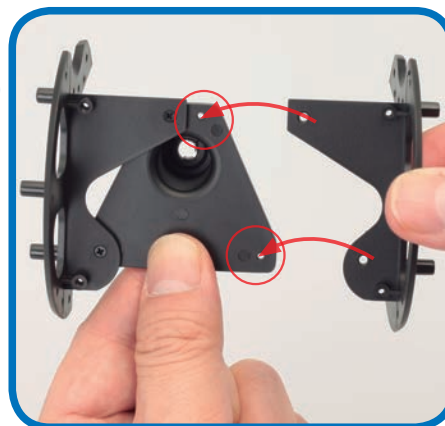
2 Take the neck mounting in your right hand and offer up the left head frame as shown. Double-check that you have selected the correct side.



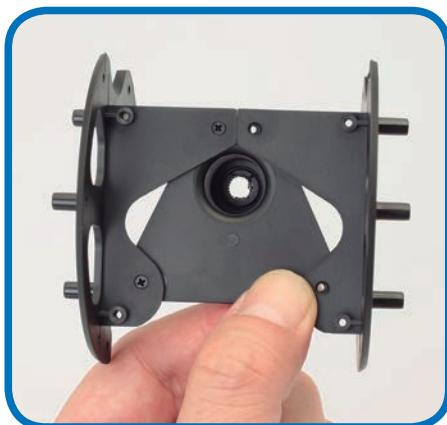
3 Place the parts together so that the two holes circled in red line up.



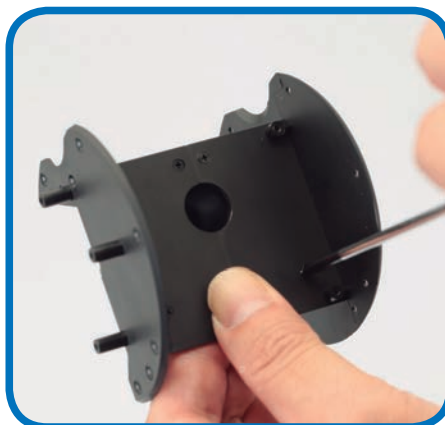
4 Fit one of the M2 x 4.5mm countersunk screws into each of the holes and tighten them with the screwdriver. Be careful not to overtighten the screws.



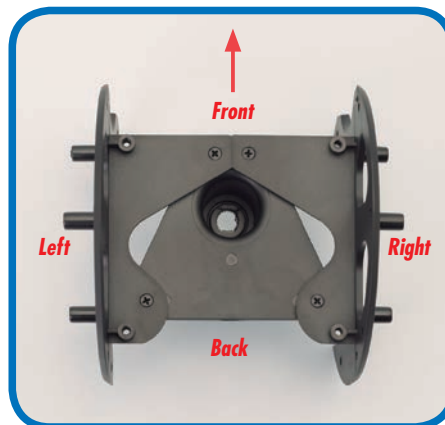
5 Now align the right head frame with the neck mounting as indicated in red, so that the holes match as they did in Step 3.



6 Holding the parts together, fit two M2 x 4.5mm countersunk screws into the circled holes.



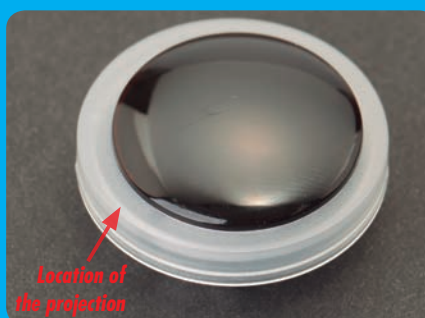
7 Tighten both screws with the screwdriver.



8 That completes the assembly of the head frames. This view from above shows which way round it will fit in the head.

ASSEMBLING ROBI'S EYES

TIP!



If you have difficulty finding the projection indicated in the photo on the left, turn the eye over. You will see that the white border is slightly wider on one side, and narrower on the other. The projection is situated directly below the widest point of the white border.

9

Lay either of the eyes assembled in Stage 1 face down on your worktop. Find the single small projection on the outer edge of the white border.



10

Take either of the silver eye sockets and look for a small notch in the silver rim, above the two screw holes.



11

Insert the eye into the socket, so that the projection circled in Step 9 fits into the notch in the rim and the parts fit together snugly.



12

Holding the eye in place, take one strip of black tape and start wrapping it around the edge of the white border, peeling away the backing paper as you go. Make sure that the tape does not overlap the front of the eye.



13

There is more than enough tape to go round once, so continue wrapping when you reach the starting point – it is not necessary to cut the tape.



14

Repeat Steps 9-13 for the second eye and eye socket. You will see that there is no difference between the left and right eyes.

FITTING ROBI'S EYES



- 15** Pick up either of the completed eye assemblies. The first one you fit will become Robi's right eye.



- 16** Hold the head framework from Step 8 as shown, turning the eye around until the circled holes in the eye frame line up with those on the right of the head frame.



- 17** Insert the eye behind the head frame, and align the holes so that they are ready to be screwed together.



- 18** Holding the parts together, insert and tighten two of the M2 x 4.5mm countersunk screws.



- 19** Repeat Steps 15-18 to attach the left eye. You may find this easier if you turn the head frame assembly upside down as shown.



- 20** Your assembly should look like this. Check that the wider part of the whites of Robi's eyes are towards the outside of his head.

FITTING ROBI'S FACE



- 21** Line up the face with the head assembly, as shown. The two curves on the upper edge fit against the edges of the eyes.



- 22** Fit the face between the left and right head frames so that the screw holes on the left and right sides of the face line up with the corresponding holes circled in red.



- 23** Insert two of the M2 x 4.5mm countersunk screws into the holes on the right side of the head, indicated in Step 22. Tighten with the screwdriver.



- 24** Turn the assembly over and repeat Step 23 on the left of the face.



Assembled head frame

That completes this stage of the assembly, with the features of Robi's face beginning to take shape. Store it carefully until next time.



In your next pack – Robi 02

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 3

Begin assembling the neck stand

PARTS PROVIDED

Components for Robi's neck stand, which will be used to test the movement of Robi's head.



Stage 4

Continue building Robi's head, and assemble your first servo motor

PARTS PROVIDED

More parts for Robi's head and neck stand, plus the first of the servo motors that enable him to move.

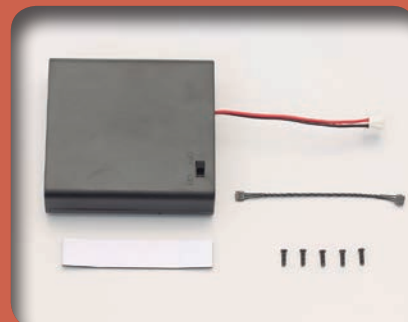


Stage 5

Complete the neck stand, and fit the servo that will move Robi's head

PARTS PROVIDED

The final components that make up the outer casing of Robi's neck stand.



Stage 6

See the first of Robi's moving parts in action, as you test his head servo

PARTS PROVIDED

The battery pack that will power Robi's servos, and another servo cable.

